Hedy Lamarr: The Mother of WiFi

By Suryatapa Saha

It is strange that women need a special platform just to make the world aware of their achievements in STEM. The recognition is necessary, and I will not pretend otherwise, yet as a woman in science, it feels odd to be treated like an endangered species whose existence must be proven through competitions like this one. Growing up, I never knew that women scientists existed at all. Every scientist in my childhood stories was a man, and the dream of becoming one felt like it belonged to someone else. Now, in the same field, I find myself competing simply to prove that I deserve the same opportunities and have the same intellectual ability. I have heard people say that more girls study biology than physics because biology is “easier,” and that women should choose the arts because they are “less demanding.” But no subject is easy, and no discipline belongs to one gender. Such remarks often come from ego and insecurity, from those who are uncomfortable seeing women succeed in STEM, or in any field at all. This is why stories like that of Hedy Lamarr matter. Born Hedwig Eva Maria Kiesler, she was an Austrian-American actress celebrated on screen, but she was also an inventor who co-developed a radio guidance system during World War II. The world remembered her face, yet it took decades to remember her mind.

I first came to know about Hedy Lamarr through an article I happened to read online. I expected it to be about an old Hollywood actress, but it said she was also an inventor. That surprised me, and I wanted to know more. I started researching her life, and I found out that she is often called the “mother of WiFi.” I felt both amazed and a little angry. Why had I never heard about her in school?

Hedy Lamarr was born in Vienna, Austria, on 9 November 1914. Her father was a bank director and her mother was a pianist. From a young age she was curious about how things worked, and she grew up loving both the stage and the world of inventions. As a young woman she began acting in European films. In 1933 she appeared in the film *Ecstasy*, which made her famous but also controversial. From then on, people talked about her looks far more than her talent.

Her personal life was not easy. She married Fritz Mandl, a wealthy arms manufacturer who was very controlling. At his dinners she sat among men who discussed weapons and military technology, and she quietly listened and learned. Later she escaped the marriage and left for Europe's film world again, where she met Louis B. Mayer, the head of the MGM studio. He took her to Hollywood and gave her the name Hedy Lamarr. She became a major star, but the studio mostly saw her as a beautiful face. Behind the glamour, she spent her free time inventing.

This is where her struggle as a woman in science truly begins. When World War II started, Hedy wanted to help. She knew that radio-controlled torpedoes could be sent off course if enemies jammed their signal. Together with the composer George Antheil, she designed a system in which the signal keeps jumping between different frequencies, so it cannot be easily blocked. This idea is called frequency hopping. They received a patent in 1942 and gave it to the US Navy. But when Hedy tried to offer her talent to the government's National Inventors Council, she was told she would be more useful raising money with her famous face. She did that too, and she raised millions of dollars. Yet her invention was left on a shelf. Her patent expired in 1959, and she never received any payment for it.

The biggest problem she faced was that people could not believe a beautiful woman could also be intelligent. Because she was called "the most beautiful woman in the world," nobody wanted to look beyond her face. Her beauty became a mask that hid her mind. This is a misconception many women still face today: that a woman can be either beautiful or smart, but not both.

Her later life was hard. Her film career slowed down, she had money problems, and she faced personal troubles, including a highly publicised arrest, that made the news forget her achievements. Meanwhile, her idea quietly became useful in the real world. The Navy later used her principle in its own systems, and frequency hopping became part of the foundation of modern wireless technology such as Bluetooth and WiFi. To be accurate, she did not invent WiFi directly, but her idea helped make the technology that connects our phones and laptops possible. Recognition came very late. In 1997 she and Antheil received the Electronic Frontier Foundation Pioneer Award, and she passed away in 2000. In 2014 she was finally inducted into the National Inventors Hall of Fame, more than seventy years after her patent.

Hedy Lamarr's story is not only hers. Many women in science go through the same thing. We work hard, but we also carry an extra weight: the weight of proving ourselves again and again, of hearing that our subject is "easy," of being doubted before we even speak. This causes a mental stress that our male colleagues rarely have to think about, though they have their own struggles too. And still, we are here. Women are studying, researching, discovering and inventing every day, in labs, in classrooms and in companies around the world. Hedy was ignored in her time, but history proved her right. Today's women in science will not wait seventy years to be recognised. Lamarr once joked, "Any girl can be glamorous. All you have to do is stand still and look stupid." She refused to stand still. So will we, and the girls who come after us will grow up knowing that scientists can look just like them.

NOTE:

If this essay has made you curious about Hedy Lamarr, I encourage you to read more about her life. She did leave an autobiography, *Ecstasy and Me*, published in 1966, but it was written with ghostwriters, and she later said it was full of lies and fought to stop it. Even that is a lesson. A woman who gave the world a groundbreaking idea had her own life story turned into a scandal she did not write. To know the real Hedy, read the careful biographies, such as Richard Rhodes's *Hedy's Folly* and Stephen Michael Shearer's *Beautiful: The Life of Hedy Lamarr*, and read the patent she filed herself. Study her life, and let her be remembered for what she created, not for how she looked. Her ideas outlasted her fame, and they deserve the same attention.

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